Python Sets

Create a Set in Python

```
# create a set of integer type
student_id = {112, 114, 116, 118, 115}
print('Student ID:', student_id)

# create a set of string type
vowel_letters = {'a', 'e', 'i', 'o', 'u'}
print('Vowel Letters:', vowel_letters)

# create a set of mixed data types
mixed_set = {'Hello', 101, -2, 'Bye'}
print('Set of mixed data types:', mixed_set)
```

```
Student ID: {112, 114, 115, 116, 118}
Vowel Letters: {'e', 'u', 'i', 'a', 'o'}
Set of mixed data types: {101, 'Bye', 'Hello', -2}
```

Create an Empty Set in Python

```
# create an empty set
empty_set = set()

# create an empty dictionary
empty_dictionary = { }

# check data type of empty_set
print('Data type of empty_set:', type(empty_set))

# check data type of dictionary_set
print('Data type of empty_dictionary:', type(empty_dictionary))
```

```
Data type of empty_set: <class 'set'>
Data type of empty_dictionary: <class 'dict'>
```

Duplicate Items in a Set

```
numbers = {2, 4, 6, 6, 2, 8}
print(numbers)   # {8, 2, 4, 6}
```

```
{8, 2, 4, 6}
```

Add and Update Set Items in Python

Add Items to a Set in Python

```
numbers = {21, 34, 54, 12}

print('Initial Set:',numbers)

# using add() method
numbers.add(32)

print('Updated Set:', numbers)
```

```
Initial Set: {34, 12, 21, 54}
Updated Set: {32, 34, 12, 21, 54}
```

Update Python Set

```
companies = {'Lacoste', 'Ralph Lauren'}
tech_companies = ['apple', 'google', 'apple']

# using update() method
companies.update(tech_companies)

print(companies)

# Output: {'google', 'apple', 'Lacoste', 'Ralph Lauren'}
```

```
{'Lacoste', 'Ralph Lauren', 'apple', 'google'}
```

Remove an Element from a Set

```
languages = {'Swift', 'Java', 'Python'}

print('Initial Set:',languages)

# remove 'Java' from a set
removedValue = languages.discard('Java')

print('Set after remove():', languages)
```

```
Initial Set: {'Swift', 'Java', 'Python'}
Set after remove(): {'Swift', 'Python'}
```

Iterate Over a Set in Python

```
fruits = {"Apple", "Peach", "Mango"}

# for loop to access each fruits
for fruit in fruits:
    print(fruit)
```

```
Peach
Apple
Mango
```

Find Number of Set Elements

```
even_numbers = {2,4,6,8}
print('Set:',even_numbers)

# find number of elements
print('Total Elements:', len(even_numbers))
```

```
Set: {8, 2, 4, 6}
Total Elements: 4
```

Python Set Operations

Union of Two Sets

```
# first set
A = {1, 3, 5}

# second set
B = {0, 2, 4}

# perform union operation using |
print('Union using |:', A | B)

# perform union operation using union()
print('Union using union():', A.union(B))
```

```
Union using |: {0, 1, 2, 3, 4, 5}
Union using union(): {0, 1, 2, 3, 4, 5}
```

Set Intersection

```
# first set
A = {1, 3, 5}

# second set
B = {1, 2, 3}

# perform intersection operation using &
print('Intersection using &:', A & B)

# perform intersection operation using intersection()
print('Intersection using intersection():', A.intersection(B))
```

```
Intersection using &: {1, 3}
Intersection using intersection(): {1, 3}
```

Difference between Two Sets

```
# first set
A = {2, 3, 5}

# second set
B = {1, 2, 6}
```

```
# perform difference operation using &
print('Difference using &:', A - B)

# perform difference operation using difference()
print('Difference using difference():', A.difference(B))
```

```
Difference using &: {3, 5}
Difference using difference(): {3, 5}
```

Set Symmetric Difference

```
# first set
A = {2, 3, 5}

# second set
B = {1, 2, 6}

# perform difference operation using &
print('using ^:', A ^ B)

# using symmetric_difference()
print('using symmetric_difference():', A.symmetric_difference(B))
```

```
using ^: {1, 3, 5, 6}
using symmetric_difference(): {1, 3, 5, 6}
```

Check if two sets are equal

```
# first set
A = {1, 3, 5}

# second set
B = {3, 5, 1}

# perform difference operation using &
if A == B:
    print('Set A and Set B are equal')
else:
    print('Set A and Set B are not equal')
```

```
Set A and Set B are equal
```